Content Determination of Sodium Monohydrogen Phosphate

Method for the alkalimetric content determination of Sodium Monohydrogen Phosphate

Sodium monohydrogen phosphate is a saline laxative which is used for emptying the bowels, alkalising urine, hypercalcaemia and in cases of hypophosphataemia.

Sample	Na-monohydrogen phosphate
Substance	Na-monohydrogen phosphate M = 142.0 g/mol
Chemicals	Deionised water R
Titrant	c(NaOH) = 0.5 mol/L c(HCl) = 0.5 mol/L
Standard	Potassium hydrogen phthalate (M521), THAM (M522)
Instruments	METTLER TOLEDO DL70ES/DL77
Accessories	Titration beaker ME-101974 SP250 Peristaltic pump Rondo60 Sample changer
Indication	DG111-SC

Chemistry

1. Protonation of the monohydrogen phosphate to phosphoric acid through hydrochloric acid

$$HPO_4^{2-} + 2H_3O^+ \rightarrow H_3PO_4 + 2H_2O$$

2. Neutralisation of the excess hydrochloric acid as well as deprotonation to secondary phosphate in the first equivalence point

$$H_3PO_4 + OH^- \rightarrow H_2PO_4^- + H_2O$$

3. Deprotonation to primary phosphate in the second equivalence point

$$H_2PO_4^- + OH^- \rightarrow HPO_4^{2-}$$

Calculation

R(%) = (QDISP-Q) • M • F/(10 • m)

QDISP= Total amount of dispensed HCl in mmol
Q = Titrant consumption in mmol
F = 100/(100-H20) Formula for the loss on drying of Na_2HPO_4
M = Molar mass Na_2HPO_4
m = Sample size in g

Structure

Literature

Ph. Eur. 1997; Monograph "Sodium Monohydrogen Phosphate"

Sample preparation

Weigh a sample of 0.24 to 0.28 g sodium monohydrogen phosphate, accurately to 0.1 mg, into a 100 mL titration beaker and dissolve in 40 mL deionised water.

Remarks

The previously determined loss on drying content is saved as the auxilliary value H20.
The curve of the back titration shows two equivalence points so that the content of sodium monohydrogen phosphate as well as sodium dihydrogen phosphate can be determined on a parallel basis.

Waste disposal

Halogen-free organic solvents waste.

Author

L. Pludra

Results

METTLER DL77 Titrator V3.1 **Dr. Kade Pharm. Fabrik GmbH**
Tante Elfriede **Quality Control Laboratory**

Method: G031 Cont. Na2HPO4 Date: 14-12-1999 13:20
User: **Pludra** Measured: 14-12-1999 15:11

RESULTS

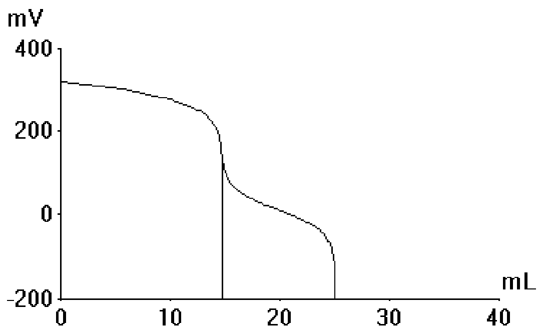
No.	ID1	Sample size and results
1/1	992223	0.9178 g R1 = 100.38 % Na2HPO4
1/2	992223	0.9477 g R1 = 99.96 % Na2HPO4
1/3	992223	1.0699 g R1 = 99.72 % Na2HPO4
1/4	992223	1.0368 g R1 = 99.87 % Na2HPO4
1/5	992223	1.0046 g R1 = 100.05 % Na2HPO4
1/6	992223	0.9735 g R1 = 99.76 % Na2HPO4
		R3 = 19.00 % Drying loss

STATISTICS
Number of results R1 $\underline{n}$ = 6
Mean value $\bar{x}$ = 99.96 % Na2HPO4
Standard deviation s = 0.238792 % Na2HPO4
Rel. standard deviation srel = 0.239 %

Table of measured values

EQP	ET	Volume mL	Signal mV	1st Deriv. mV/mL	Time min:s
	ET1	0.0000	320.3		0:03
		5.7140	301.0	-3.4	0:13
		8.5700	285.5	-5.4	0:20
	ET2	10.0000	275.7	-6.9	0:26
		10.2000	273.8	-9.5	0:29
		10.4000	272.1	-8.8	0:33
		10.6000	270.4	-8.4	0:36
		10.8000	268.5	-9.5	0:40
		11.0000	267.0	-7.4	0:44
		\\	\\	\\	\\
		14.5440	173.9	-148.1	2:10
		14.5880	165.2	-198.9	2:19
		14.6200	158.0	-225.3	2:25
		14.6500	150.7	-242.7	2:31
EQP1		14.6800	142.3	-280.0	2:38
		14.7040	136.8	-227.5	2:45
		14.7480	128.5	-189.3	2:51
		14.8000	119.5	-173.7	2:59
		\\	\\	\\	\\
		24.4040	-62.4	-44.8	6:53
		24.5400	-68.9	-48.4	6:57
		24.6920	-80.1	-73.2	7:02
		24.7720	-87.1	-87.5	7:08
		24.8500	-96.9	-126.5	7:14
		24.8980	-105.1	-170.6	7:20
		24.9340	-112.1	-192.5	7:26
		24.9700	-119.8	-213.9	7:33
EQP2		25.0040	-127.9	-238.8	7:39
		25.0340	-134.9	-233.3	7:45
		25.0680	-142.2	-216.2	7:52

Titration curve



Method

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Method:     G031     Cont. Na2HPO4
             Version:  14-12-1999  13:20

Title
   Method ID  . . . . . . . . . . . . . G031
   Title  . . . . . . . . . . . . . . . Cont. Na2HPO4
   Date/time  . . . . . . . . . . . . . 14-Dec-1999  13:20
Sample
   Number samples . . . . . . . . . . . 6
   Titration stand  . . . . . . . . . . ST20 1
   Entry type . . . . . . . . . . . . . Weight m
      Lower limit [g] . . . . . . . . . 0.9
      Upper limit [g] . . . . . . . . . 1.1
   ID1  . . . . . . . . . . . . . . . .
   Molar mass M . . . . . . . . . . . . 100.0
   Equivalent number z  . . . . . . . . 1
   Temperature sensor . . . . . . . . . Manual
Dispense
   Titrant  . . . . . . . . . . . . . . HCl
   Concentration [mol/L]  . . . . . . . 0.5
   Volume [mL]  . . . . . . . . . . . . 25
Stir
   Speed [%]  . . . . . . . . . . . . . 50
   Time [s] . . . . . . . . . . . . . . 10
Titration
   Titrant  . . . . . . . . . . . . . . NaOH
   Concentration [mol/L]  . . . . . . . 0.5
   Sensor . . . . . . . . . . . . . . . DG111-SC
   Unit of meas.  . . . . . . . . . . . mV
   Titration mode . . . . . . . . . . . EQP
      Predispensing 1 . . . . . . . . . mL
         Volume [mL]  . . . . . . . . . 10
      Titrant addition  . . . . . . . . DYN
         dE(set) [mV] . . . . . . . . . 8.0
         Limits dV  . . . . . . . . . . Absolute
            dV(min) [mL]  . . . . . . . 0.02
            dV(max) [mL]  . . . . . . . 0.2
      Measure mode  . . . . . . . . . . EQU
         dE [mV]  . . . . . . . . . . . 0.5
         dt [s] . . . . . . . . . . . . 1.0
         t(min) [s] . . . . . . . . . . 3.0
         t(max) [s] . . . . . . . . . . 30.0
      Threshold . . . . . . . . . . . . 100.0
      Maximum volume [mL] . . . . . . . 30.0
      Termination after n EQPs  . . . . Yes
         n =  . . . . . . . . . . . . . 2
      Evaluation procedure  . . . . . . Standard
      Stop for reevaluation . . . . . . Yes
         Condition  . . . . . . . . . . neq<2
Calculation
   Result name  . . . . . . . . . . . . Na2HPO4
   Formula  . . . . . . . . . . . . . . R1=1420*(QDISP-Q1)/C
   Constant . . . . . . . . . . . . . . C=m*(100-H20)
   Result unit  . . . . . . . . . . . . %
   Decimal places . . . . . . . . . . . 2
Calculation
   Result name  . . . . . . . . . . . . NaH2PO4
   Formula  . . . . . . . . . . . . . . R2=(Q1+Q2-QDISP)/C2
   Constant . . . . . . . . . . . . . . C2=(QDISP-Q1)/100
   Result unit  . . . . . . . . . . . . %
   Decimal places . . . . . . . . . . . 2
   Condition  . . . . . . . . . . . . . Yes
      Condition . . . . . . . . . . . . Q1+Q2>QDISP
Rinse
   Auxiliary reagent  . . . . . . . . . Water
   Volume [mL]  . . . . . . . . . . . . 30.0
Record
   Output unit  . . . . . . . . . . . . Computer
   Raw results last sample  . . . . . . Yes
   Table of values  . . . . . . . . . . Yes
   E - V curve  . . . . . . . . . . . . Yes
Statistics
   Ri (i=index) . . . . . . . . . . . . R1
   Standard deviation s . . . . . . . . Yes
   Rel. standard deviation srel . . . . Yes
Statistics
   Ri (i=index) . . . . . . . . . . . . R2
   Standard deviation s . . . . . . . . Yes
   Rel. standard deviation srel . . . . Yes
Calculation
   Result name  . . . . . . . . . . . . Loss on drying
   Formula  . . . . . . . . . . . . . . R3=H20
   Constant . . . . . . . . . . . . . .
   Result unit  . . . . . . . . . . . . %
   Decimal places . . . . . . . . . . . 2
Record
   Output unit  . . . . . . . . . . . . Printer
   All results  . . . . . . . . . . . . Yes
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